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Nafe Abubaker

GitHub
LinkedIn
Portfolio

Professional Experience

AI Engineer Intern **Sep 2026 – Present**
Cellula Technologies ***Remote***

- Built and evaluated RNN and LSTM models for nine-class NLP classification using TensorFlow/Keras, combining user queries with image captions.
- Developed a RAG pipeline using PDF chunking, sentence embeddings, FAISS retrieval, and LLM generation; deployed NLP and document Q&A apps with Streamlit.

Data Science & Machine Learning Trainee **Jan 2026 – Apr 2026**
IBT x GGateway **Remote**

- Progressed through an intensive end-to-end Data Science & ML bootcamp, from Python, SQL, data cleaning, EDA, and statistics to machine learning, deep learning, NLP, LLMs, and deployment.
- Applied the full workflow in a team capstone on student dropout prediction, contributing to EDA, data preparation, modeling, evaluation, and presentation; earned the bootcamp's top final evaluation.

Backend Engineering Trainee **Oct 2024 – Jan 2025**
ASAL Technologies **On-site (Rawabi)**

- Developed RESTful APIs in C#/.NET Core with MongoDB CRUD operations.
- Implemented authentication, validation, and collaborated with cross-functional teams on backend integration.

Data Engineering Trainee **Feb 2025 – Jun 2025**
ASAL Technologies **On-site (Rawabi)**

- Built an event-driven Prefect 2.0 ETL pipeline triggered by MongoDB receipt inserts, calculating operational KPIs and idempotently upserting results into four MySQL analytics tables.
- Exposed four analytics datasets through asynchronous FastAPI endpoints using SQLAlchemy and Pydantic, covering receipt summaries, monthly store sales, item performance, and peak hours.

Education

Birzeit University B.Sc. in Computer Engineering – GPA: 3.2/*4.0* **Ramallah** **2020 – 2025**

Projects

Student Dropout Prediction [GitHub] **Apr 2026**

- Coordinated the project and led data and EDA work for a recall-prioritized dropout-risk model using the HSLS:09 dataset of approximately 15,900 students.
- Compared Logistic Regression, Random Forest, XGBoost, and LightGBM; a tuned XGBoost model at a 0.30 threshold achieved 86.5% recall, identifying 423 of 489 dropouts in the test set
- **Technologies**: Python, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, Matplotlib, Seaborn.

Traffic Management & Red-Light Violation [GitHub] **Jul 2025**

- Built an end-to-end computer vision system for queue-aware traffic signal recommendations using overhead drone footage (YOLOv8 + SORT + polygon ROIs).
- Achieved 91.7% mAP@0.5 on the validation set for YOLOv8 vehicle detection, and improved traffic throughput by ~0.25 cars/s compared to fixed timing (simulation-based).
- Implemented red-light violation detection using stop-line logic and best-frame evidence capture, then extracted license plates using plate detection + OCR.
- **Technologies**: YOLOv8, SORT, OpenCV, OCR, Roboflow, FastAPI, React, MongoDB.

Arabic Question Answering [GitHub] **Jan 2024**

- Built a simple Arabic Q&A demo using a pre-trained AraELECTRA question-answering model.
- Applied AraBERT-style preprocessing (ArabertPreprocessor) and tokenization to format question/context inputs and extract answer spans.
- **Technologies**: PyTorch, Transformers, AraBERT, NLP.

Palestine Store Hub [GitHub] **Jan 2025**

- Developed 21 REST endpoints across three ASP.NET Core controllers for users, stores, and products, supporting CRUD, search, filtering, geolocation-based discovery, ratings, and recommendations
- Integrated three MongoDB collections through dedicated service providers, implementing duplicate checks, partial updates, image validation, Swagger documentation, and CORS-based frontend integration.
- **Technologies**: C#/.NET Core, MongoDB, REST APIs.